uOttawa

Seismic Risk Assessment (CVG 6320)

Assignment 02

Submitted by:

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Submitted to

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A) Determine the probabilities of collapse associated with complete damage state for both cases.

Case 1: Conventional construction of moment resisting frames (assume all frames are identical in geometry and stiffness).

Step 1: Determine the seismic region from Range and Median MCEr Response (Table 5-1 FEMA P 155)

Table 5-1 Range and Median MCE_R Spectral Response Acceleration Values in Each Seismicity Region

Seismicity Region	Range of Response Values for Each Region		Median Response Values for Each Region	
	S_S (g)	S_1 (g)	$S_{S,avg}$ (g)	$S_{1,avg}$ (g)
Low (L)	$S_S < 0.250g$	$S_1 < 0.100g$	0.20	0.08
Moderate (M)	$0.250g \leq S_S < 0.500g$	$0.100g \leq S_1 < 0.200g$	0.40	0.16
Moderately High (MH)	$0.500g \leq S_S < 1.000g$	$0.200g \leq S_1 < 0.400g$	0.80	0.32
High (H)	$1.000g \leq S_S < 1.500g$	$0.400g \leq S_1 < 0.600g$	1.20	0.48
Very High (VH)	$S_S \geq 1.500g$	$S_1 \geq 0.600g$	2.25	0.90

Storey Height	3.6
Building type	Conventional construction of moment resisting frames
Site class	C
City	Ottawa
No of storey	6
βc,p	0.75

The spectral acceleration values is provided by the NBCC 2020 Seismic Hazard Tool (https://www.seismescanada.rncan.gc.ca/hazard-alea/interpolat/nbc2020- cnb2020-en.php) for Ottawa, ON.

Sa(0.2) [g]	Sa(0.5) [g]	Sa(1.0) [g]	Sa(2.0) [g]	Sa(5.0) [g]	Sa(10.0) [g]	PGA [g]	PGV [m/s]
0.698	0.411	0.218	0.0993	0.0261	0.00856	0.377	0.282

So, comparing the results from data to table 5-1 in FEMA P155, Seismicity region of the city is Moderately high (MH).

Step 2: Determine the building type;

From the table 3-3 FEMA P155, The designated building type is C1.

FEMA Building Type	Photograph
C1 Concrete moment-resisting frames	

Step 3: Development of the Capacity Curve

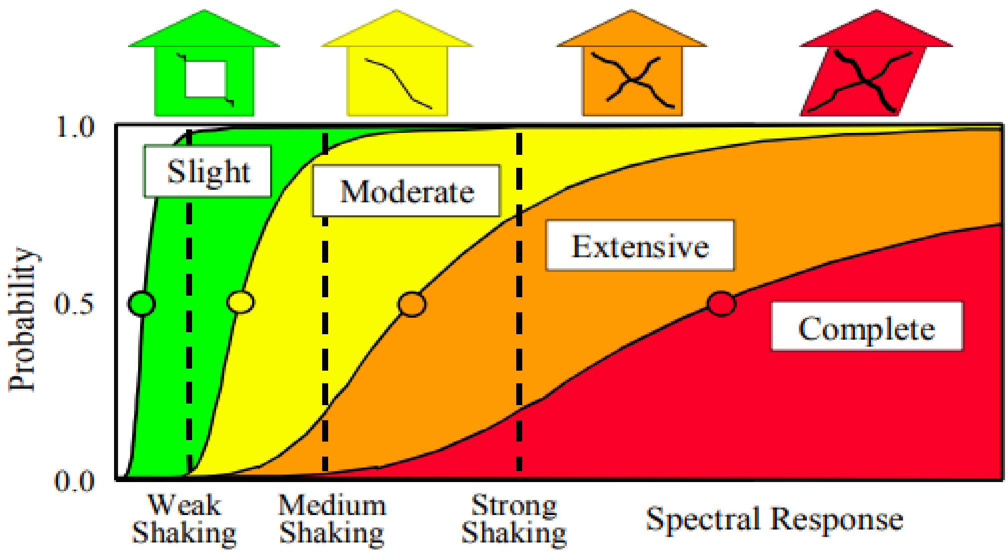


Figure 4-5 Example fragility curves for Slight, Moderate, Extensive and Complete structural damage.

From FEMA P155 4.4.2.1, Capacity curve is used to plot the lateral load resistance of the building as a function of lateral displacement. The demand spectra represent what can be considered as weak, medium, and strong ground shaking, and the two building capacity curves represent weaker and stronger construction. So, the demand spectra the lower the shaking of the building and lower the probability of the collapse.

To calculate the capacity curve following data are obtained from table A-1 to A-11 from FEMA P155.

HRbuilding height, ft	60
Te elastic period, seconds	0.89
Cs, Seismic design coefficient	0.043
γ, yield strength factor	1.8
λ, overstrength factor	2

m ductility factor	3.82
α1 modal weight factor	0.73
α2 modal height factor	0.72
α3 modal shape factor	1.62
bE elastic damping	7
K, degradation factor	0.4
Δc storey drift ratio	0.035
bC,DLognormal standard deviation	0.92
Collapse factor	0.13

The building capacity curve is assumed to be linear when the spectral displacement is less than the yield displacement and is assumed to remain 5-10 5: Development of Third Edition Basic Scores and Modifiers FEMA P-155 plastic past the ultimate point. The transition from yield point to ultimate point of the capacity curve is assumed to be elliptical with the following form (from Equation 4-2):

$$\frac{(D-D_u)^2}{a^2}+\frac{(A-k)^2}{b^2}=1$$

Where, a, b, and k are constants with the following values from Equations 4-3 through 4-5:

$$a=\sqrt{\frac{D_y}{A_y}b^2\frac{(D_u-D_y)}{(A_y-k)}}\,;$$

$$b=A_u-k$$

$$k=\frac{A_u^2-A_y^2+\frac{A_y^2}{D_y}(D_y-D_u)}{2(A_u-A_y)+\frac{A_y}{D_y}(D_y-D_u)}$$

To get this values we first have to calculate the ultimate point and yield point which are as below:

Yield Point:
Ay = Csγ/α1
Dy = 9.8 AyTe2

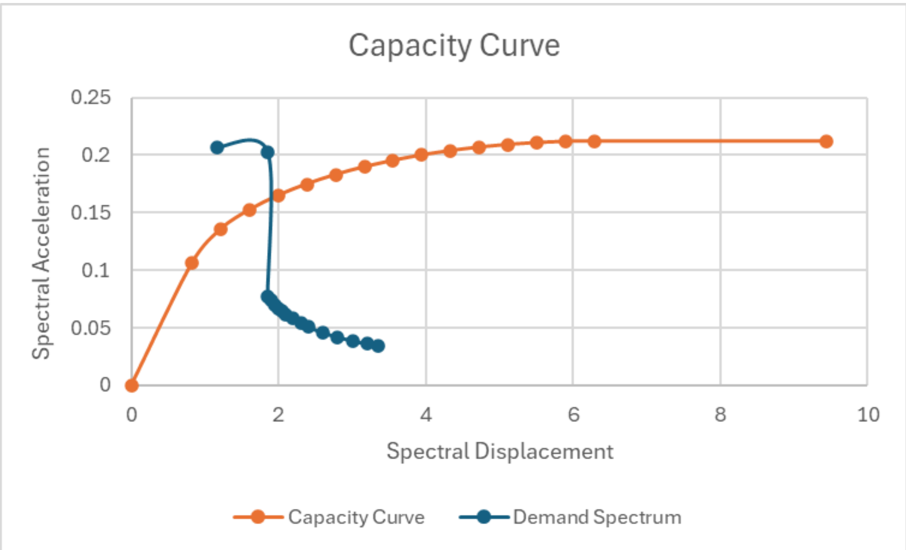
Ultimate Point:
Au = λAy
Du = λμDy

Yied points	
Ay	0.1060
Dy	0.8230
Ultimate Points	

Au	0.2121
Du	6.2881
a	5.5530
b	0.1289
k	0.0832

Obtain D via interval between Dy and Du and obtain the corresponding A values and tabulate the demand spectrum values:

D	A	Moment Resisting Frame	
		Spectral Acceleration Sa	Spectral Displacement Sd
		(g)	(in)
0	0		0
0.8230	0.1060	0.2070	1.1590
1.2134	0.1355	0.2030	1.8560
1.6038	0.1524	0.0770	1.8560
1.9941	0.1649	0.0740	1.9020
2.3845	0.1748	0.0700	1.9510
2.7748	0.1830	0.0670	2.0010
3.1652	0.1897	0.0650	2.0550
3.5556	0.1954	0.0620	2.1030
3.9459	0.2000	0.0580	2.2020
4.3363	0.2038	0.0540	2.3020
4.7266	0.2069	0.0510	2.4030
5.1170	0.2092	0.0460	2.6010
5.5074	0.2108	0.0420	2.8010
5.8977	0.2117	0.0390	3.0030
6.2881	0.2121	0.0360	3.2020
9.4321	0.2121	0.0340	3.3520



It can be said that, when capacity curve graph is compared to the fig 4-5 in FEMA P155, it falls under the medium shaking and after ultimate points the spectral displacement stays constant with the spectral acceleration.

Step 4: Determination of Peak Response

The two curves intersect at a spectral displacement of D = 1.8560 in. This is the peak response of the building for the given seismic demand.

Step 5: Determination of Probability of Complete Damage

To calculate the probability of complete damage by using equation 4-10 in FEMA P-155.

P [Complete Damage] = $\varphi[(1/\beta C,P) \ln (D/Sd,c)]$

Sd,c = Δc*Hr*($\alpha 2/\alpha 3$)

The lognormal standard deviation (beta) factor is 0.75.

Sd,c	11.2
D	1.856
P[Complete Damage]	-2.39665
P[Complete Damage] with Standard deviation	0.008273

Step 6: Determination of Probability of collaps

P[Collapse] = Collapse factor*P [Complete Damage]

Proability of Collaps	0.001075
	0.107546

Hence, the probabilities of Complete damage for moment resisting frame is 0.108%.

Case 2: Conventional Shear walls

Step 1: Determine the seismic region from Range and Median MCEr Response (Table 5-1 FEMA P 155)

Table 5-1 Range and Median MCE_R Spectral Response Acceleration Values in Each Seismicity Region

Seismicity Region	Range of Response Values for Each Region		Median Response Values for Each Region	
	S_S (g)	S_1 (g)	$S_{S,avg}$ (g)	$S_{1,avg}$ (g)
Low (L)	$S_S < 0.250g$	$S_1 < 0.100g$	0.20	0.08
Moderate (M)	$0.250g \le S_S < 0.500g$	$0.100g \le S_1 < 0.200g$	0.40	0.16
Moderately High (MH)	$0.500g \le S_S < 1.000g$	$0.200g \le S_1 < 0.400g$	0.80	0.32
High (H)	$1.000g \le S_S < 1.500g$	$0.400g \le S_1 < 0.600g$	1.20	0.48
Very High (VH)	$S_S \ge 1.500g$	$S_1 \ge 0.600g$	2.25	0.90

Storey Height	3.6
Building type	Conventional Shear Wall
Site class	C
City	Ottawa
No of storey	6
$\beta c,p$	0.75

The spectral acceleration values is provided by the NBCC 2020 Seismic Hazard Tool (https://www.seismescanada.rncan.gc.ca/hazard-alea/interpolat/nbc2020- cnb2020-en.php) for Ottawa, ON.

Sa(0.2) [g]	Sa(0.5) [g]	Sa(1.0) [g]	Sa(2.0) [g]	Sa(5.0) [g]	Sa(10.0) [g]	PGA [g]	PGV [m/s]
0.698	0.411	0.218	0.0993	0.0261	0.00856	0.377	0.282

So, comparing the results from data to table 5-1 in FEMA P155, Seismicity region of the city is Moderately high (MH).

Step 2: Determine the building type;

From the table 3-3 FEMA P155, The designated building type is C2.

C2 Concrete shear wall buildings	

Step 3: Development of the Capacity Curve

From FEMA P155 4.4.2.1, Capacity curve is used to plot the lateral load resistance of the building as a function of lateral displacement. The demand spectra represent what can be considered as weak, medium, and strong ground shaking, and the two building capacity curves represent weaker and stronger construction. So, lower the demand spectra the lower the shaking of the building and lower the probability of the collapse but the building can also have the low probability of total damage with entensive shaking due to building design and construction quality.

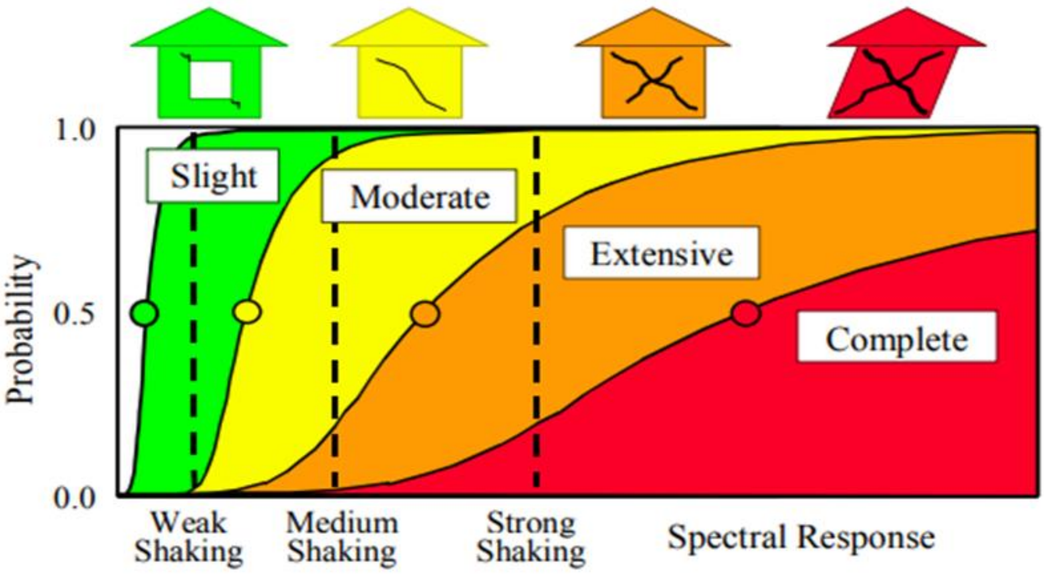


Figure 4-5 Example fragility curves for Slight, Moderate, Extensive and Complete structural damage.

To calculate the capacity curve the following data are obtained from table A-1 to A-11 from FEMA P155.

HRbuilding height, ft	60
Te elastic period, seconds	0.65
Cs, Seismic design coefficient	0.043
γ, yield strength factor	1.8
λ, overstrength factor	2
m ductility factor	3.82
α1 modal weight factor	0.79
α2 modal height factor	0.72
α3 modal shape factor	1.62
bE elastic damping	7
K, degradation factor	0.4
Δc storey drift ratio	0.055
bC,DLognormal standard deviation	0.92
Collapse factor	0.13

The building capacity curve is assumed to be linear when the spectral displacement is less than the yield displacement and is assumed to remain 5-10 5: Development of Third Edition Basic Scores and Modifiers FEMA P-155 plastic past the ultimate point. The transition from yield point to ultimate point of the capacity curve is assumed to be elliptical with the following form (from Equation 4-2):

$$\frac{(D-D_u)^2}{a^2}+\frac{(A-k)^2}{b^2}=1$$

Where, a, b, and k are constants with the following values from Equations 4-3 through 4-5:

$$a=\sqrt{\frac{D_y}{A_y}b^2\frac{(D_u-D_y)}{(A_y-k)}}\,;$$

$$b=A_u-k$$

$$k=\frac{A_u^2-A_y^2+\frac{A_y^2}{D_y}(D_y-D_u)}{2(A_u-A_y)+\frac{A_y}{D_y}(D_y-D_u)}$$

To get this values we first have to calculate the ultimate point and yield point which are as below:

Yield Point:

Ay = Csγ/α1

Dy = 9.8 AyTe2

Ultimate Point:

Au = λAy

Du = λμDy

Yied points	

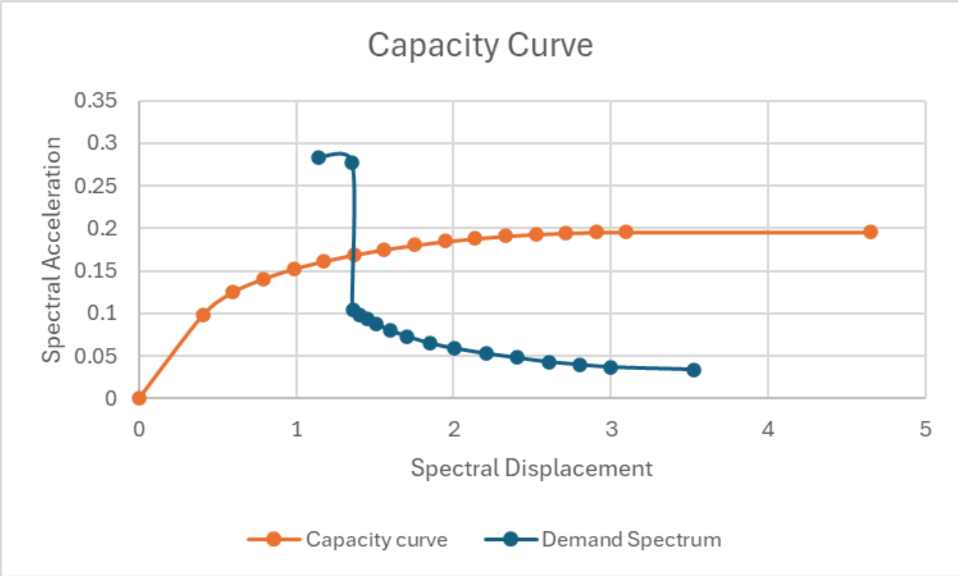
Ay	0.0980
Dy	0.4057
Ultimate Points	
Au	0.1959
Du	3.0993
a	2.7370
b	0.1191
k	0.0769

Obtain D via interval between Dy and Du and obtain the corresponding A values and tabulate the demand spectrum values:

After finding D for equal intervals we have find corresponding A values with following formula:

$A = \sqrt{1} - (D - \mathrm{Du}) \div \mathrm{a}^\wedge 2 \ + \mathrm{k}$

D	A	Moment Resisting Frame	
		Spectral Acceleration Sa	Spectral Displacement Sd
		(g)	(in)
0	0		0
0.4057	0.0980	0.2830	1.1390
0.5981	0.1252	0.2780	1.3560
0.7905	0.1408	0.1050	1.3580
0.9829	0.1524	0.0990	1.4000
1.1753	0.1616	0.0940	1.4500
1.3677	0.1691	0.0880	1.5060
1.5601	0.1753	0.0800	1.6000
1.7525	0.1805	0.0730	1.7060
1.9449	0.1848	0.0650	1.8520
2.1373	0.1884	0.0590	2.0080
2.3297	0.1911	0.0530	2.2040
2.5221	0.1933	0.0480	2.4020
2.7145	0.1948	0.0430	2.6060
2.9069	0.1957	0.0400	2.8000
3.0993	0.1959	0.0370	3.0010
4.6489	0.1959	0.0340	3.5270



It can be said that, when capacity curve graph is compared to the fig 4-5 in FEMA P155, it falls under the Extensive shaking and after ultimate points the spectral displacement stays constant with the spectral acceleration.

Step 4: Determination of Peak Response

The two curves intersect at a spectral displacement of D = 1.3580 in. This is the peak response of the building for the given seismic demand.

Step 5: Determination of Probability of Complete Damage

To calculate the probability of complete damage by using equation 4-10 in FEMA P-155.

P [Complete Damage] = $\varphi[(1/\beta C,P) \ln (D/Sd,c)]$

$Sd,c = \Delta c * Hr * (\alpha 2/\alpha 3)$

The lognormal standard deviation (beta) factor is 0.75.

Sd,c	17.6
D	1.358
P[Complete Damage]	-3.41585
	0.000318

Step 6: Determination of Probability of collaps

P[Collapse] = Collapse factor*P [Complete Damage]

Proability of Collaps	4.13E-05
	0.004133

Hence, the probabilities of Complete damage for moment resisting frame is 0.0041%.

B) Determine the probabilities of collapse associated with complete damage state for both cases if the buildings were constructed in 1915 (Pre-Code).

Case 1: Conventional construction of moment resisting frames (assume all frames are identical in geometry and stiffness).

Step 1: Determine the seismic region from Range and Median MCEr Response (Table 5-1 FEMA P 155)

Storey Height	3.6
Building type	Conventional construction of moment resisting frames
Site class	C
City	Ottawa
No of storey	6
$\beta c,p$	0.75

The spectral acceleration values is provided by the NBCC 2020 Seismic Hazard Tool (https://www.seismescanada.rncan.gc.ca/hazard-alea/interpolat/nbc2020- cnb2020-en.php) for Ottawa, ON.

Sa(0.2) [g]	Sa(0.5) [g]	Sa(1.0) [g]	Sa(2.0) [g]	Sa(5.0) [g]	Sa(10.0) [g]	PGA [g]	PGV [m/s]
0.698	0.411	0.218	0.0993	0.0261	0.00856	0.377	0.282

So, comparing the results from data to table 5-1 in FEMA P155, Seismicity region of the city is Moderately high (MH).

Step 2: Determine the building type;

From the table 3-3 FEMA P155, The designated building type is C1

Step 3: Development of the Capacity Curve
To calculate the capacity curve following data are obtained from table A-1 to A-11 from FEMA P155.

HRbuilding height, ft	60
Te elastic period, seconds	0.89
Cs, Seismic design coefficient	0.029
γ, yield strength factor	1.8
λ, overstrength factor	1.5
m ductility factor	3.82
α1 modal weight factor	0.73
α2 modal height factor	0.72
α3 modal shape factor	1.62
bE elastic damping	7
K, degradation factor	0.4
Δc storey drift ratio	0.035
bC,DLognormal standard deviation	0.97

Collapse factor	0.13

The building capacity curve is assumed to be linear when the spectral displacement is less than the yield displacement and is assumed to remain 5-10 5: Development of Third Edition Basic Scores and Modifiers FEMA P-155 plastic past the ultimate point. The transition from yield point to ultimate point of the capacity curve is assumed to be elliptical with the following form (from Equation 4-2):

$$\frac{(D-D_u)^2}{a^2}+\frac{(A-k)^2}{b^2}=1$$

Where, a, b, and k are constants with the following values from Equations 4-3 through 4-5:

$$a=\sqrt{\frac{D_y}{A_y}b^2\frac{(D_u-D_y)}{(A_y-k)}}:$$

$$b=A_u-k$$

$$k=\frac{A_u^2-A_y^2+\frac{A_y^2}{D_y}(D_y-D_u)}{2(A_u-A_y)+\frac{A_y}{D_y}(D_y-D_u)}$$

To get this values we first have to calculate the ultimate point and yield point which are as below:

Yield Point:
Ay = Csγ/α1
Dy = 9.8 AyTe2

Ultimate Point:
Au = λAy
Du = λμDy

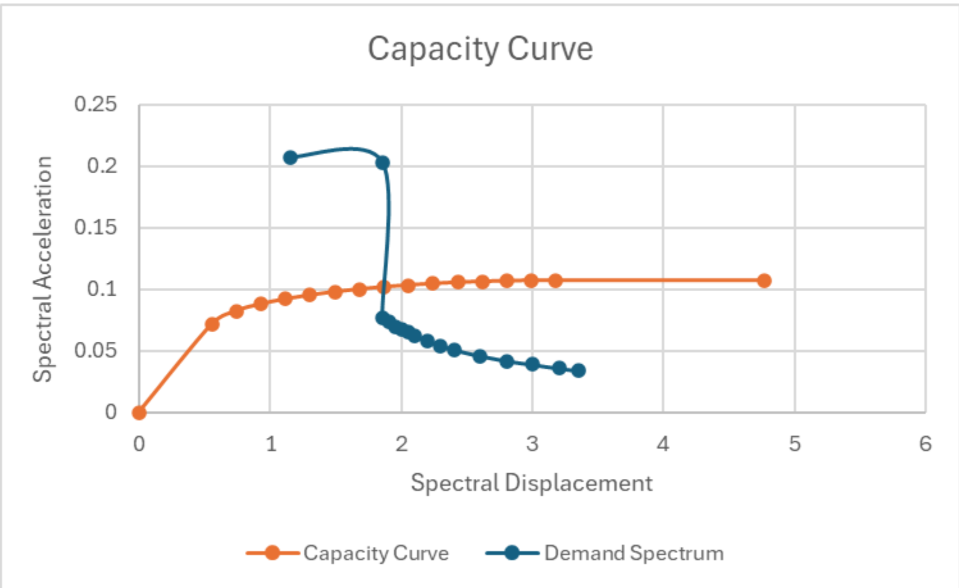
Yied points	
Ay	0.0715
Dy	0.5551
Ultimate Points	
Au	0.1073
Du	3.1806
a	2.6441
b	0.0405
k	0.0667

Obtain D via interval between Dy and Du and obtain the corresponding A values and tabulate the demand spectrum values:

After finding D for equal intervals we have find corresponding A values with following formula:

$A = \sqrt{1} - (D - \mathrm{Du}) \div \mathrm{a}^2 \ + \mathrm{k}$

D	A	Moment Resisting Frame	
		Spectral Acceleration Sa	Spectral Displacement Sd
		(g)	(in)
0	0		0
0.5551	0.0715	0.2070	1.1590
0.7426	0.0824	0.2030	1.8560
0.9302	0.0880	0.0770	1.8560
1.1177	0.0921	0.0740	1.9020
1.3052	0.0953	0.0700	1.9510
1.4928	0.0979	0.0670	2.0010
1.6803	0.1001	0.0650	2.0550
1.8678	0.1019	0.0620	2.1030
2.0554	0.1034	0.0580	2.2020
2.2429	0.1046	0.0540	2.3020
2.4304	0.1056	0.0510	2.4030
2.6180	0.1063	0.0460	2.6010
2.8055	0.1069	0.0420	2.8010
2.9931	0.1072	0.0390	3.0030
3.1806	0.1073	0.0360	3.2020
4.7709	0.1073	0.0340	3.3520



Step 4: Determination of Peak Response

The two curves intersect at a spectral displacement of D = 1.8560 in. This is the peak response of the building for the given seismic demand.

Step 5: Determination of Probability of Complete Damage

To calculate the probability of complete damage by using equation 4-10 in FEMA P-155.

P [Complete Damage] = $\varphi[(1/\beta C,P) \ln (D/Sd,c)]$

$Sd,c = \Delta c*Hr*(\alpha 2/\alpha 3)$

The lognormal standard deviation (beta) factor is 0.75.

Sd,c	11.2
D	1.856
P[Complete Damage]	-2.39665
	0.008273

Step 6: Determination of Probability of collaps

P[Collapse] = Collapse factor*P [Complete Damage]

Proability of Collaps	0.001075
	0.107546

Hence, the probabilities of Complete damage for moment resisting frame is 0.108%.

Case 2: Conventional Shear walls

Step 1: Determine the seismic region from Range and Median MCEr Response (Table 5-1 FEMA P 155)

Storey Height	3.6
Building type	Conventional Shear Wall
Site class	C
City	Ottawa
No of storey	6
βc,p	0.75

The spectral acceleration values is provided by the NBCC 2020 Seismic Hazard Tool (https://www.seismescanada.rncan.gc.ca/hazard-alea/interpolat/nbc2020- cnb2020-en.php) for Ottawa, ON.

Sa(0.2) [g]	Sa(0.5) [g]	Sa(1.0) [g]	Sa(2.0) [g]	Sa(5.0) [g]	Sa(10.0) [g]	PGA [g]	PGV [m/s]
0.698	0.411	0.218	0.0993	0.0261	0.00856	0.377	0.282

So, comparing the results from data to table 5-1 in FEMA P155, Seismicity region of the city is

Moderately high (MH).

Step 2: Determine the building type;

From the table 3-3 FEMA P155, The designated building type is C2

Step 3: Development of the Capacity Curve
To calculate the capacity curve following data are obtained from table A-1 to A-11 from FEMA P155.

HRbuilding height, ft	60
Te elastic period, seconds	0.65
Cs, Seismic design coefficient	0.029
γ, yield strength factor	1.8
λ, overstrength factor	1.5
m ductility factor	3.82
α1 modal weight factor	0.79
α2 modal height factor	0.72
α3 modal shape factor	1.62
bE elastic damping	7
K, degradation factor	0.4
Δc storey drift ratio	0.05
bC,DLognormal standard deviation	0.97
Collapse factor	0.13

The building capacity curve is assumed to be linear when the spectral displacement is less than the yield displacement and is assumed to remain 5-10 5: Development of Third Edition Basic Scores and Modifiers FEMA P-155 plastic past the ultimate point. The transition from yield point to ultimate point of the capacity curve is assumed to be elliptical with the following form (from Equation 4-2):

$$\frac{(D-D_u)^2}{a^2}+\frac{(A-k)^2}{b^2}=1$$

Where, a, b, and k are constants with the following values from Equations 4-3 through 4-5:

$$a=\sqrt{\frac{D_y}{A_y}b^2\frac{(D_u-D_y)}{(A_y-k)}}:$$

$$b=A_u-k$$

$$k=\frac{A_u^2-A_y^2+\frac{A_y^2}{D_y}(D_y-D_u)}{2(A_u-A_y)+\frac{A_y}{D_y}(D_y-D_u)}$$

To get this values we first have to calculate the ultimate point and yield point which are as below:

Yield Point:

$Ay = Cs\gamma/\alpha 1$
$Dy = 9.8\ AyTe2$

Ultimate Point:
$Au = \lambda Ay$
$Du = \lambda\mu Dy$

Yied points	
Ay	0.0661
Dy	0.2736
Ultimate Points	
Au	0.0991
Du	1.5677
a	1.3032
b	0.0375
k	0.0616

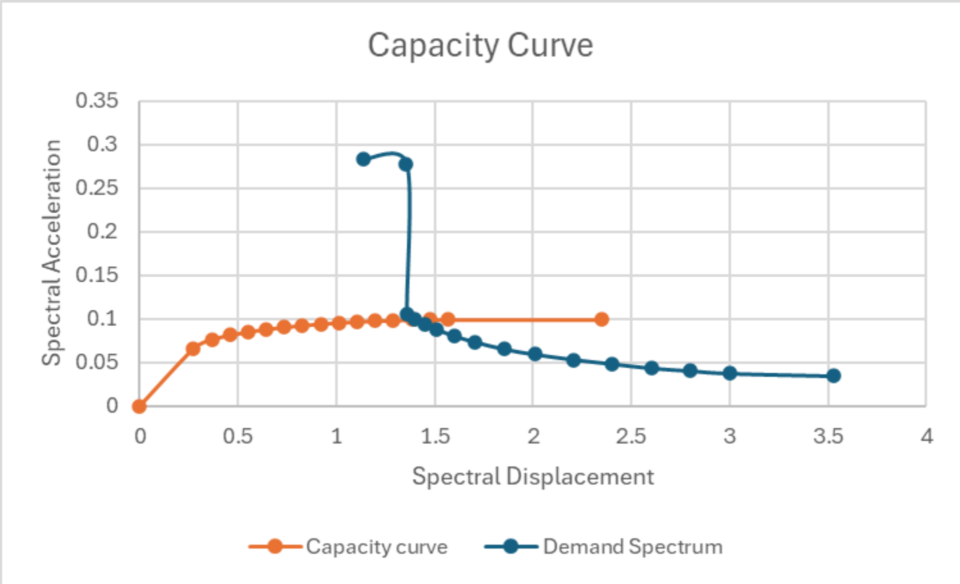
Obtain D via interval between Dy and Du and obtain the corresponding A values and tabulate the demand spectrum values:

After finding D for equal intervals we have find corresponding A values with following formula:

$$A = \sqrt{1} - (D - \mathrm{Du}) \div \mathrm{a}^\wedge 2\ + \mathrm{k}$$

D	A	Moment Resisting Frame	
		Spectral Acceleration Sa	Spectral Displacement Sd
		(g)	(in)
0	0		0
0.2736	0.0661	0.2830	1.1390
0.3660	0.0761	0.2780	1.3560
0.4585	0.0813	0.1050	1.3580
0.5509	0.0851	0.0990	1.4000
0.6433	0.0881	0.0940	1.4500
0.7358	0.0905	0.0880	1.5060
0.8282	0.0925	0.0800	1.6000
0.9206	0.0942	0.0730	1.7060
1.0131	0.0956	0.0650	1.8520
1.1055	0.0967	0.0590	2.0080
1.1979	0.0976	0.0530	2.2040
1.2904	0.0983	0.0480	2.4020
1.3828	0.0987	0.0430	2.6060
1.4752	0.0990	0.0400	2.8000

1.5677	0.0991	0.0370	3.0010
2.3515	0.0991	0.0340	3.5270



Step 4: Determination of Peak Response

The two curves intersect at a spectral displacement of D = 1.358 in. This is the peak response of the building for the given seismic demand.

Step 5: Determination of Probability of Complete Damage

To calculate the probability of complete damage by using equation 4-10 in FEMA P-155.

P [Complete Damage] = φ[(1/βC,P) ln (D/Sd,c)]

Sd,c = Δc*Hr*(α2/α3)

The lognormal standard deviation (beta) factor is 0.75.

Sd,c	16
D	1.358
P[Complete Damage]	-3.28877
	0.000503

Step 6: Determination of Probability of collaps

P[Collapse] = Collapse factor*P [Complete Damage]

Proability of Collaps	6.54E-05
	0.006541

Hence, the probabilities of Complete damage for moment resisting frame is 0.0065%.

When pre code is compared with basic score, the Probability of complete damage for moment resisting frame is almost similar. However, Probability of complete damage for shear wall is higher in pre code than basic score.

C) Determine the probabilities of collapse associated with complete damage state for both cases if the buildings were constructed in 2010 (Post-Benchmark).

Case 1: Conventional construction of moment resisting frames (assume all frames are identical in geometry and stiffness).

Step 1: Determine the seismic region from Range and Median MCEr Response (Table 5-1 FEMA P 155)

Storey Height	3.6
Building type	Conventional construction of moment resisting frames
Site class	C
City	Ottawa
No of storey	6
βc,p	0.75

The spectral acceleration values is provided by the NBCC 2020 Seismic Hazard Tool (https://www.seismescanada.rncan.gc.ca/hazard-alea/interpolat/nbc2020- cnb2020-en.php) for Ottawa, ON.

Sa(0.2) [g]	Sa(0.5) [g]	Sa(1.0) [g]	Sa(2.0) [g]	Sa(5.0) [g]	Sa(10.0) [g]	PGA [g]	PGV [m/s]
0.698	0.411	0.218	0.0993	0.0261	0.00856	0.377	0.282

So, comparing the results from data to table 5-1 in FEMA P155, Seismicity region of the city is Moderately high (MH).

Step 2: Determine the building type;

From the table 3-3 FEMA P155, The designated building type is C1

Step 3: Development of the Capacity Curve
To calculate the capacity curve following data are obtained from table A-1 to A-11 from FEMA P155.

HRbuilding height, ft	60
Te elastic period, seconds	0.89
Cs, Seismic design coefficient	0.107
γ, yield strength factor	1.8
λ, overstrength factor	2
m ductility factor	5.08
α1 modal weight factor	0.73
α2 modal height factor	0.72
α3 modal shape factor	1.62
bE elastic damping	7

K, degradation factor	0.6
Δc storey drift ratio	0.083
bC,DLognormal standard deviation	0.82
Collapse factor	0.13

The building capacity curve is assumed to be linear when the spectral displacement is less than the yield displacement and is assumed to remain 5-10 5: Development of Third Edition Basic Scores and Modifiers FEMA P-155 plastic past the ultimate point. The transition from yield point to ultimate point of the capacity curve is assumed to be elliptical with the following form (from Equation 4-2):

$$\frac{(D-D_u)^2}{a^2}+\frac{(A-k)^2}{b^2}=1$$

Where, a, b, and k are constants with the following values from Equations 4-3 through 4-5:

$$a=\sqrt{\frac{D_y}{A_y}b^2\frac{(D_u-D_y)}{(A_y-k)}};$$

$$b=A_u-k$$

$$k=\frac{A_u^2-A_y^2+\frac{A_y^2}{D_y}(D_y-D_u)}{2(A_u-A_y)+\frac{A_y}{D_y}(D_y-D_u)}$$

To get this values we first have to calculate the ultimate point and yield point which are as below:

Yield Point:
Ay = Csγ/α1
Dy = 9.8 AyTe2

Ultimate Point:
Au = λAy
Du = λμDy

Yied points	
Ay	0.2638
Dy	2.0480
Ultimate Points	
Au	0.5277
Du	20.8081
a	18.9026

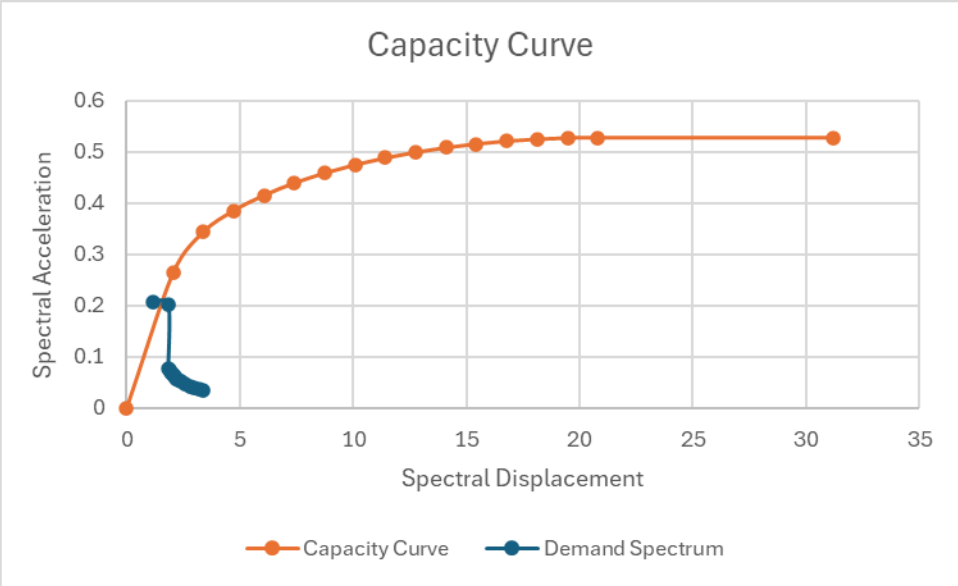
b	0.3007
k	0.2270

Obtain D via interval between Dy and Du and obtain the corresponding A values and tabulate the demand spectrum values:

After finding D for equal intervals we have find corresponding A values with following formula:

$A = \sqrt{1} - (D - \mathrm{Du}) \div \mathrm{a}^\wedge 2\ + \mathrm{k}$

D	A	Moment Resisting Frame	
		Spectral Acceleration Sa	Spectral Displacement Sd
		(g)	(in)
0	0		0
2.0480	0.2638	0.2070	1.1590
3.3881	0.3437	0.2030	1.8560
4.7281	0.3851	0.0770	1.8560
6.0681	0.4152	0.0740	1.9020
7.4081	0.4391	0.0700	1.9510
8.7481	0.4585	0.0670	2.0010
10.0881	0.4746	0.0650	2.0550
11.4281	0.4880	0.0620	2.1030
12.7681	0.4991	0.0580	2.2020
14.1081	0.5081	0.0540	2.3020
15.4481	0.5153	0.0510	2.4030
16.7881	0.5208	0.0460	2.6010
18.1281	0.5246	0.0420	2.8010
19.4681	0.5269	0.0390	3.0030
20.8081	0.5277	0.0360	3.2020
31.2122	0.5277	0.0340	3.3520



Step 4: Determination of Peak Response

The two curves intersect at a spectral displacement of D = 1.80 in. This is the peak response of the building for the given seismic demand.

Step 5: Determination of Probability of Complete Damage

To calculate the probability of complete damage by using equation 4-10 in FEMA P-155.

P [Complete Damage] = $\varphi[(1/\beta_{C,P}) \ln (D/S_{d,c})]$

$S_{d,c} = \Delta_c * H_r * (\alpha_2/\alpha_3)$

The lognormal standard deviation (beta) factor is 0.75.

Sd,c	26.56
D	1.8
P[Complete Damage]	-3.58883
	0.000166

Step 6: Determination of Probability of collaps

P[Collapse] = Collapse factor*P [Complete Damage]

Proability of Collaps	2.16E-05
	0.002159

Hence, the probabilities of Complete damage for moment resisting frame is 0.0021%.

Case 2: Conventional Shear walls

Step 1: Determine the seismic region from Range and Median MCEr Response (Table 5-1 FEMA P 155)

Storey Height	3.6
Building type	Conventional Shear Wall
Site class	C
City	Ottawa
No of storey	6
$\beta c,p$	0.75

The spectral acceleration values is provided by the NBCC 2020 Seismic Hazard Tool (https://www.seismescanada.rncan.gc.ca/hazard-alea/interpolat/nbc2020- cnb2020-en.php) for Ottawa, ON.

Sa(0.2) [g]	Sa(0.5) [g]	Sa(1.0) [g]	Sa(2.0) [g]	Sa(5.0) [g]	Sa(10.0) [g]	PGA [g]	PGV [m/s]
0.698	0.411	0.218	0.0993	0.0261	0.00856	0.377	0.282

So, comparing the results from data to table 5-1 in FEMA P155, Seismicity region of the city is Moderately high (MH).

Step 2: Determine the building type;

From the table 3-3 FEMA P155, The designated building type is C2

Step 3: Development of the Capacity Curve

To calculate the capacity curve following data are obtained from table A-1 to A-11 from FEMA P155.

HRbuilding height, ft	60
Te elastic period, seconds	0.65
Cs, Seismic design coefficient	0.107
γ, yield strength factor	1.8
λ, overstrength factor	2
m ductility factor	5.08
$\alpha 1$ modal weight factor	0.79
$\alpha 2$ modal height factor	0.72
$\alpha 3$ modal shape factor	1.62
bE elastic damping	7
K, degradation factor	0.6
Δc storey drift ratio	0.083
bC,DLognormal standard deviation	0.82
Collapse factor	0.13

The building capacity curve is assumed to be linear when the spectral displacement is less than the yield displacement and is assumed to remain 5-10 5: Development of Third Edition Basic Scores and Modifiers FEMA P-155 plastic past the ultimate point. The transition from yield point to ultimate point of the capacity curve is assumed to be elliptical with the following form (from Equation 4-2):

$$\frac{(D-D_u)^2}{a^2}+\frac{(A-k)^2}{b^2}=1$$

Where, a, b, and k are constants with the following values from Equations 4-3 through 4-5:

$$a=\sqrt{\frac{D_y}{A_y}b^2\,\frac{(D_u-D_y)}{(A_y-k)}}\,;$$

$$b=A_u-k$$

$$k=\frac{A_u^2-A_y^2+\dfrac{A_y^2}{D_y}(D_y-D_u)}{2(A_u-A_y)+\dfrac{A_y}{D_y}(D_y-D_u)}$$

To get this values we first have to calculate the ultimate point and yield point which are as below:

Yield Point:
Ay = Csγ/α1
Dy = 9.8 AyTe2

Ultimate Point:
Au = λAy
Du = λμDy

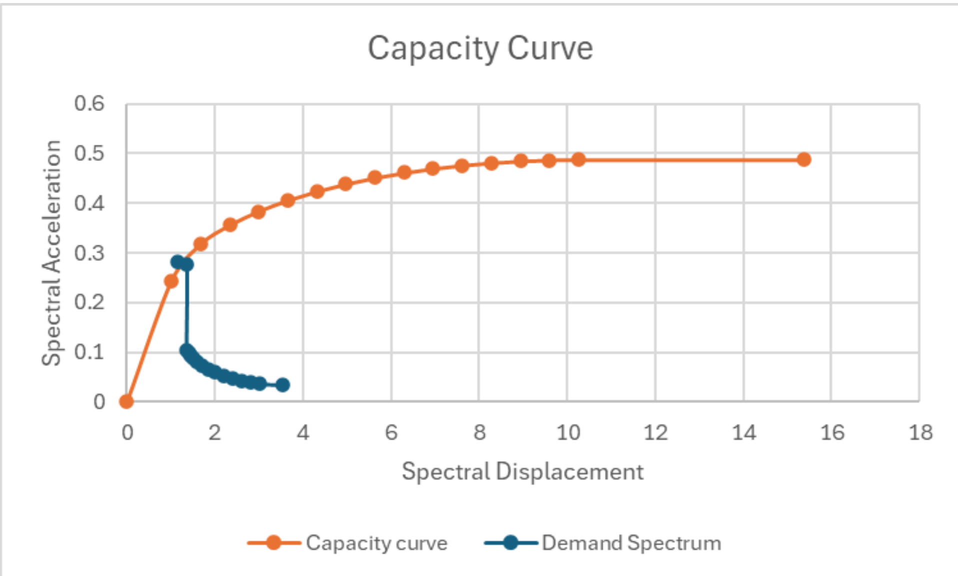
Yied points	
Ay	0.2438
Dy	1.0094
Ultimate Points	
Au	0.4876
Du	10.2559
a	9.3167
b	0.2778
k	0.2097

Obtain D via interval between Dy and Du and obtain the corresponding A values and tabulate the demand spectrum values:

After finding D for equal intervals we have find corresponding A values with following formula:

$$A=\sqrt{1}-(D-\mathrm{Du})\div \mathrm{a}^\wedge 2\ +\mathrm{k}$$

D	A	Moment Resisting Frame	
		Spectral Acceleration Sa	Spectral Displacement Sd
		(g)	(in)
0	0		0
1.0094	0.2438	0.2830	1.1390
1.6699	0.3176	0.2780	1.3560
2.3304	0.3558	0.1050	1.3580
2.9908	0.3837	0.0990	1.4000
3.6513	0.4057	0.0940	1.4500
4.3118	0.4237	0.0880	1.5060
4.9722	0.4386	0.0800	1.6000
5.6327	0.4510	0.0730	1.7060
6.2932	0.4612	0.0650	1.8520
6.9536	0.4696	0.0590	2.0080
7.6141	0.4762	0.0530	2.2040
8.2746	0.4812	0.0480	2.4020
8.9350	0.4848	0.0430	2.6060
9.5955	0.4869	0.0400	2.8000
10.2559	0.4876	0.0370	3.0010
15.3839	0.4876	0.0340	3.5270



Step 4: Determination of Peak Response

The two curves intersect at a spectral displacement of D = 1.30 in. This is the peak response of the building for the given seismic demand.

Step 5: Determination of Probability of Complete Damage

To calculate the probability of complete damage by using equation 4-10 in FEMA P-155.

P [Complete Damage] = $\varphi[(1/\beta_{C,P}) \ln (D/S_{d,c})]$

$S_{d,c} = \Delta_c * H_r * (\alpha_2/\alpha_3)$

The lognormal standard deviation (beta) factor is 0.75.

Sd,c	26.56
D	1.3
P[Complete Damage]	-4.02272
	2.88E-05

Step 6: Determination of Probability of collaps

P[Collapse] = Collapse factor*P [Complete Damage]

Probability of Collaps	3.74E-06
	0.000374

Hence, the probability of Complete damage for moment resisting frame is 0.00037%.

The Probability of complete damage for moment resisting frame and shear wall in post benchmark is very low (almost zero) when compared to basic score.

D) Determine the probabilities of collapse associated with complete damage state for both cases if the first storey height is 5.5 m and the storey height of the other storeys is 3.5 m (severe vertical irregularity).

Case 1: Conventional construction of moment resisting frames (assume all frames are identical in geometry and stiffness).

Step 1: Determine the seismic region from Range and Median MCEr Response (Table 5-1 FEMA P 155)

Storey Height	3.5
Building type	Conventional construction of moment resisting frames
Site class	C
City	Ottawa
No of storey	6
βc,p	0.75

The spectral acceleration values is provided by the NBCC 2020 Seismic Hazard Tool (https://www.seismescanada.rncan.gc.ca/hazard-alea/interpolat/nbc2020- cnb2020-en.php) for Ottawa, ON.

Sa(0.2) [g]	Sa(0.5) [g]	Sa(1.0) [g]	Sa(2.0) [g]	Sa(5.0) [g]	Sa(10.0) [g]	PGA [g]	PGV [m/s]
0.698	0.411	0.218	0.0993	0.0261	0.00856	0.377	0.282

So, comparing the results from data to table 5-1 in FEMA P155, Seismicity region of the city is Moderately high (MH).

Step 2: Determine the building type;

From the table 3-3 FEMA P155, The designated building type is C1

Step 3: Development of the Capacity Curve
To calculate the capacity curve following data are obtained from table A-1 to A-11 from FEMA P155.

HRbuilding height, ft	60
Te elastic period, seconds	0.89
Cs, Seismic design coefficient	0.043
γ, yield strength factor	1.8
λ, overstrength factor	2
m ductility factor	3.82
α1 modal weight factor	0.73
α2 modal height factor	0.72
α3 modal shape factor	4
bE elastic damping	7
K, degradation factor	0.4
Δc storey drift ratio	0.028
bC,DLognormal standard deviation	0.92
Collapse factor	0.5

The building capacity curve is assumed to be linear when the spectral displacement is less than the yield displacement and is assumed to remain 5-10 5: Development of Third Edition Basic Scores and Modifiers FEMA P-155 plastic past the ultimate point. The transition from yield point to ultimate point of the capacity curve is assumed to be elliptical with the following form (from Equation 4-2):

$$\frac{(D-D_u)^2}{a^2}+\frac{(A-k)^2}{b^2}=1$$

Where, a, b, and k are constants with the following values from Equations 4-3 through 4-5:

$$a=\sqrt{\frac{D_y}{A_y}b^2\frac{(D_u-D_y)}{(A_y-k)}}:$$

$$b=A_u-k$$

$$k = \frac{A_u^2 - A_y^2 + \frac{A_y^2}{D_y}(D_y - D_u)}{2(A_u - A_y) + \frac{A_y}{D_y}(D_y - D_u)}$$

To get this values we first have to calculate the ultimate point and yield point which are as below:

Yield Point:
Ay = Csγ/α1
Dy = 9.8 AyTe2

Ultimate Point:
Au = λAy
Du = λμDy

Yied points	
Ay	0.1060
Dy	0.8230
Ultimate Points	
Au	0.2121
Du	6.2881
a	5.5530
b	0.1289
k	0.0832

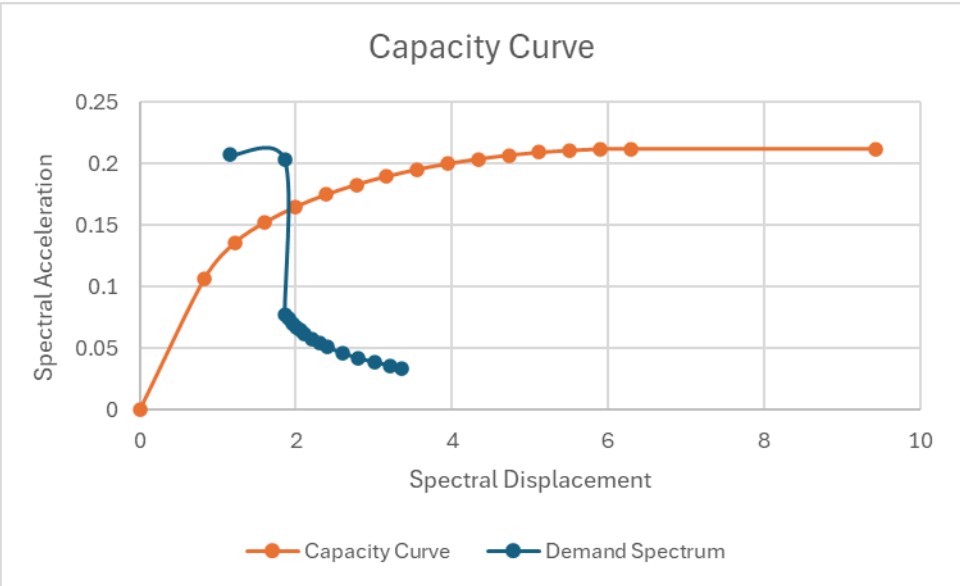
Obtain D via interval between Dy and Du and obtain the corresponding A values and tabulate the demand spectrum values:

After finding D for equal intervals we have find corresponding A values with following formula:

$A = \sqrt{1} - (D - \mathrm{Du}) \div \mathrm{a}^\wedge 2\ + \mathrm{k}$

D	A	Moment Resisting Frame	
		Spectral Acceleration Sa	Spectral Displacement Sd
		(g)	(in)
0	0		0
0.8230	0.1060	0.2070	1.1590
1.2134	0.1355	0.2030	1.8560
1.6038	0.1524	0.0770	1.8560

1.9941	0.1649	0.0740	1.9020
2.3845	0.1748	0.0700	1.9510
2.7748	0.1830	0.0670	2.0010
3.1652	0.1897	0.0650	2.0550
3.5556	0.1954	0.0620	2.1030
3.9459	0.2000	0.0580	2.2020
4.3363	0.2038	0.0540	2.3020
4.7266	0.2069	0.0510	2.4030
5.1170	0.2092	0.0460	2.6010
5.5074	0.2108	0.0420	2.8010
5.8977	0.2117	0.0390	3.0030
6.2881	0.2121	0.0360	3.2020
9.4321	0.2121	0.0340	3.3520



Step 4: Determination of Peak Response

The two curves intersect at a spectral displacement of D = 1.8560 in. This is the peak response of the building for the given seismic demand.

Step 5: Determination of Probability of Complete Damage

To calculate the probability of complete damage by using equation 4-10 in FEMA P-155.

P [Complete Damage] = $\varphi[(1/\beta C,P) \ln (D/Sd,c)]$

$Sd,c = \Delta c * Hr * (\alpha 2/\alpha 3)$

The lognormal standard deviation (beta) factor is 0.75.

Sd,c	3.6288

D	1.856
P[Complete Damage]	-0.89397
	0.185669

Step 6: Determination of Probability of collaps

P[Collapse] = Collapse factor*P [Complete Damage]

Proability of Collaps	0.092834
	9.283433

Hence, the probabilities of Complete damage for moment resisting frame is 9.283%.

Case 2: Conventional Shear walls

Step 1: Determine the seismic region from Range and Median MCEr Response (Table 5-1 FEMA P 155)

Storey Height	3.5
Building type	Conventional Shear Wall
Site class	C
City	Ottawa
No of storey	6
βc,p	0.75

The spectral acceleration values is provided by the NBCC 2020 Seismic Hazard Tool (https://www.seismescanada.rncan.gc.ca/hazard-alea/interpolat/nbc2020- cnb2020-en.php) for Ottawa, ON.

Sa(0.2) [g]	Sa(0.5) [g]	Sa(1.0) [g]	Sa(2.0) [g]	Sa(5.0) [g]	Sa(10.0) [g]	PGA [g]	PGV [m/s]
0.698	0.411	0.218	0.0993	0.0261	0.00856	0.377	0.282

So, comparing the results from data to table 5-1 in FEMA P155, Seismicity region of the city is Moderately high (MH).

Step 2: Determine the building type;

From the table 3-3 FEMA P155, The designated building type is C2

Step 3: Development of the Capacity Curve

To calculate the capacity curve following data are obtained from table A-1 to A-11 from FEMA P155.

HRbuilding height, ft	60
Te elastic period, seconds	0.65

Cs, Seismic design coefficient	0.043
γ, yield strength factor	1.8
λ, overstrength factor	2
m ductility factor	3.82
α1 modal weight factor	0.79
α2 modal height factor	0.72
α3 modal shape factor	4
bE elastic damping	7
K, degradation factor	0.4
Δc storey drift ratio	0.045
bC,DLognormal standard deviation	0.92
Collapse factor	0.5

The building capacity curve is assumed to be linear when the spectral displacement is less than the yield displacement and is assumed to remain 5-10 5: Development of Third Edition Basic Scores and Modifiers FEMA P-155 plastic past the ultimate point. The transition from yield point to ultimate point of the capacity curve is assumed to be elliptical with the following form (from Equation 4-2):

$$\frac{(D-D_u)^2}{a^2}+\frac{(A-k)^2}{b^2}=1$$

Where, a, b, and k are constants with the following values from Equations 4-3 through 4-5:

$$a=\sqrt{\frac{D_y}{A_y}b^2\frac{(D_u-D_y)}{(A_y-k)}};$$

$$b=A_u-k$$

$$k=\frac{A_u^2-A_y^2+\dfrac{A_y^2}{D_y}(D_y-D_u)}{2(A_u-A_y)+\dfrac{A_y}{D_y}(D_y-D_u)}$$

To get this values we first have to calculate the ultimate point and yield point which are as below:

Yield Point:
Ay = Csγ/α1
Dy = 9.8 AyTe2

Ultimate Point:
Au = λAy
Du = λμDy

Yied points	
Ay	0.0980
Dy	0.4057

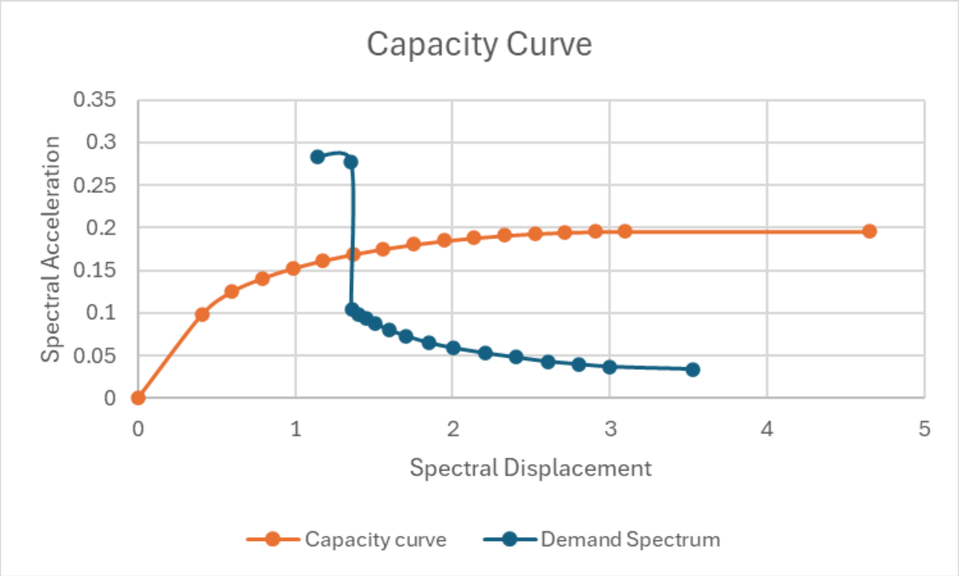
Ultimate Points	
Au	0.1959
Du	3.0993
a	2.7370
b	0.1191
k	0.0769

Obtain D via interval between Dy and Du and obtain the corresponding A values and tabulate the demand spectrum values:

After finding D for equal intervals we have find corresponding A values with following formula:

$A = \sqrt{1} - (D - \mathrm{Du}) \div \mathrm{a}^\wedge 2\ + \mathrm{k}$

D	A	Moment Resisting Frame	
		Spectral Acceleration Sa	Spectral Displacement Sd
		(g)	(in)
0	0		0
0.4057	0.0980	0.2830	1.1390
0.5981	0.1252	0.2780	1.3560
0.7905	0.1408	0.1050	1.3580
0.9829	0.1524	0.0990	1.4000
1.1753	0.1616	0.0940	1.4500
1.3677	0.1691	0.0880	1.5060
1.5601	0.1753	0.0800	1.6000
1.7525	0.1805	0.0730	1.7060
1.9449	0.1848	0.0650	1.8520
2.1373	0.1884	0.0590	2.0080
2.3297	0.1911	0.0530	2.2040
2.5221	0.1933	0.0480	2.4020
2.7145	0.1948	0.0430	2.6060
2.9069	0.1957	0.0400	2.8000
3.0993	0.1959	0.0370	3.0010
4.6489	0.1959	0.0340	3.5270



Step 4: Determination of Peak Response

The two curves intersect at a spectral displacement of D = 1.358 in. This is the peak response of the building for the given seismic demand.

Step 5: Determination of Probability of Complete Damage

To calculate the probability of complete damage by using equation 4-10 in FEMA P-155.

P [Complete Damage] = φ[(1/βC,P) ln (D/Sd,c)]

Sd,c = Δc*Hr*(α2/α3)

The lognormal standard deviation (beta) factor is 0.75.

Sd,c	5.832
D	1.358
P[Complete Damage]	- 1.94313
	0.026

Step 6: Determination of Probability of collaps

P[Collapse] = Collapse factor*P [Complete Damage]

Proability of Collaps	0.013
	1.300013

Hence, the probability of Complete damage for moment resisting frame is 1.300%.